

# Unauthorised Network Access Detection and Response Procedure

COIT13236 | Security monitoring and incident response

SEC-01 | Version 1.0 | 9 September 2026

Owner: IT Administrator | Approval: \_\_\_\_\_ | Effective date: \_\_\_\_\_

Detect and investigate unauthorised network access, suspicious remote sessions and unexpected administrative activity. The IT Administrator uses this procedure for pfSense, network devices, servers and monitored workstations.

Operating baseline: approve the proposed thresholds and response targets before activation. Review each semester and after a significant incident or system change.

## Monitoring requirements

Forward relevant pfSense firewall and system logs to the central monitoring service. Collect remote-access authentication, DHCP or device inventory records, and server login records where available. Log relevant firewall rules explicitly. Traffic between devices on the same network segment may not pass through pfSense; use endpoint or switch evidence for that activity.

| Detection condition                                                                          | Initial severity                |
|----------------------------------------------------------------------------------------------|---------------------------------|
| 10 blocked attempts from one internal device to protected services in 5 minutes              | Medium                          |
| 5 failed remote or administrative logins for one account in 5 minutes                        | High                            |
| Successful administrative access from an unapproved source, or confirmed unauthorised access | High                            |
| Active lateral movement, widespread compromise or destructive activity                       | Critical                        |
| One blocked connection or newly observed device without other indicators                     | Low; validate against inventory |

These are proposed starting thresholds. A blocked connection or unfamiliar IP alone does not prove compromise. Correlate source, destination, account, asset ownership and approved changes; use SEC-05 for authentication correlation.

## Responsibilities

The monitoring service records and routes detections. The IT Administrator owns triage and containment. The nominated backup administrator takes over missed acknowledgements under SEC-03. The project lead approves wider service disruption and accepts residual risk.

## Investigation and response

1. Open an incident record, record event and receipt times, rule ID and severity, and acknowledge the alert under SEC-03. Preserve the original event before changing the rule or host.
2. Validate the source using DHCP lease time, inventory, switch port and login evidence. Account for shared IP addresses, NAT and clock differences. Check approved maintenance with the device owner using a known contact.
3. Review related firewall, authentication and endpoint events around the detection. Determine whether traffic was blocked or allowed, whether login succeeded, which systems were reached and whether privileges changed.
4. If activity is ongoing, isolate the affected endpoint or block the narrowest confirmed malicious connection. Record the time, reason, approver and rollback method. Preserve volatile information when feasible without delaying urgent containment.
5. Use SEC-05 for compromised accounts and SEC-02 for affected shares. Avoid disabling shared infrastructure solely because a threshold fired. Escalate broader isolation to the project lead; record emergency action immediately.
6. Export relevant logs and configuration evidence using SEC-04. Remove the cause through credential recovery, access correction or host remediation. Test the change before restoring access.
7. Restore connectivity in stages and observe for at least 24 hours. Confirm normal authorised access and no recurrence. Record impact, root cause, actions, evidence and follow-up owner; the IT Administrator closes, with project lead review for High or Critical incidents.

## Notification and fallback

Send High and Critical detections immediately through SEC-03, including source IP, destination, port, firewall action and related user where known. If the monitoring or email service is unavailable, use the backup contact channel and preserve local logs for later reconciliation.

## Validation and operating record

### Acceptance tests

| Controlled test                                                                | Expected result                                                           |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| From a test endpoint, attempt access to an explicitly blocked test service     | Connection denied; matching firewall event reaches the collector          |
| Generate the configured repeat threshold against the test service              | One correlated alert with correct count, source, destination and severity |
| Generate an approved test administrative login from a non-approved test source | High alert; approval context can be recorded without deleting evidence    |
| Simulate a missed administrator acknowledgement                                | Backup escalation occurs within SEC-03 target                             |
| Restore test endpoint access after containment                                 | Authorised traffic works; monitoring remains active                       |

Run only on approved lab assets during a scheduled test window. Record event time, collector time, notification time, acknowledgement, actual result, evidence link and tester. Tests pass only when all expected results are demonstrated; retest failures.

### Routine review

Each working day, review unresolved alerts and source health. Weekly, compare unfamiliar devices with inventory and check rule exceptions. After changes, repeat a controlled detection and delivery test. Every exception must have an owner, reason, scope and expiry.

### Incident record fields

Incident ID: \_\_\_\_\_ Owner: \_\_\_\_\_ Severity: \_\_\_\_\_  
Detected and acknowledged: \_\_\_\_\_ Source and destination: \_\_\_\_\_  
Evidence location: \_\_\_\_\_ Containment and rollback: \_\_\_\_\_  
Recovery test: \_\_\_\_\_ Closure reviewer and date: \_\_\_\_\_

### References

[1] Netgate. Remote Logging with Syslog.

<https://docs.netgate.com/pfsense/en/latest/monitoring/logs/remote.html> Accessed 9 September 2026.

## Screenshot evidence appendix

These figures record selected laboratory observations. They support the tested functions described in this artefact. Proposed policies, untested controls and organisational approval remain separate from the pictured results.

Figure E1 Logged management block

Firewall / Rules / Edit

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.  
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

EMPLOYEE\_WIFI\_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

EMPLOYEE\_WIFI\_VLAN subnets

Source Address

/

Destination

Destination

☐ Invert match

Address or Alias

MANAGEMENT\_NETWORK

/

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

R06 - Block Employee WIFI Access to Management

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

☒ Display Advanced

Rule Information

Tracking ID

1787829015

Created

8/27/26 11:10:15 by admin@10.10.10.102 (Local Database)

Updated

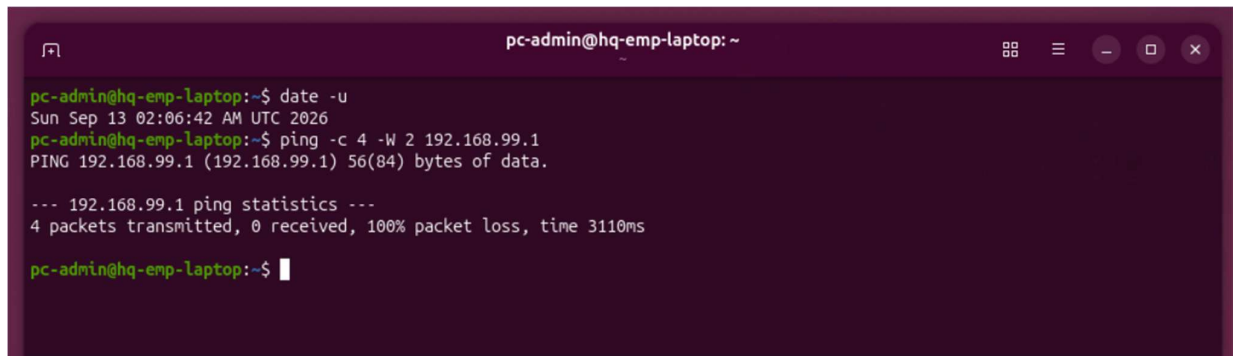
8/27/26 11:10:15 by admin@10.10.10.102 (Local Database)

The rule configuration shows logging enabled and tracker 1787829015, supporting correlation with centrally collected events.

Source evidence: evidence/HQ\_R06\_Logging\_Tracking\_ID.png

## Screenshot evidence continued

Figure E2 Controlled management access test



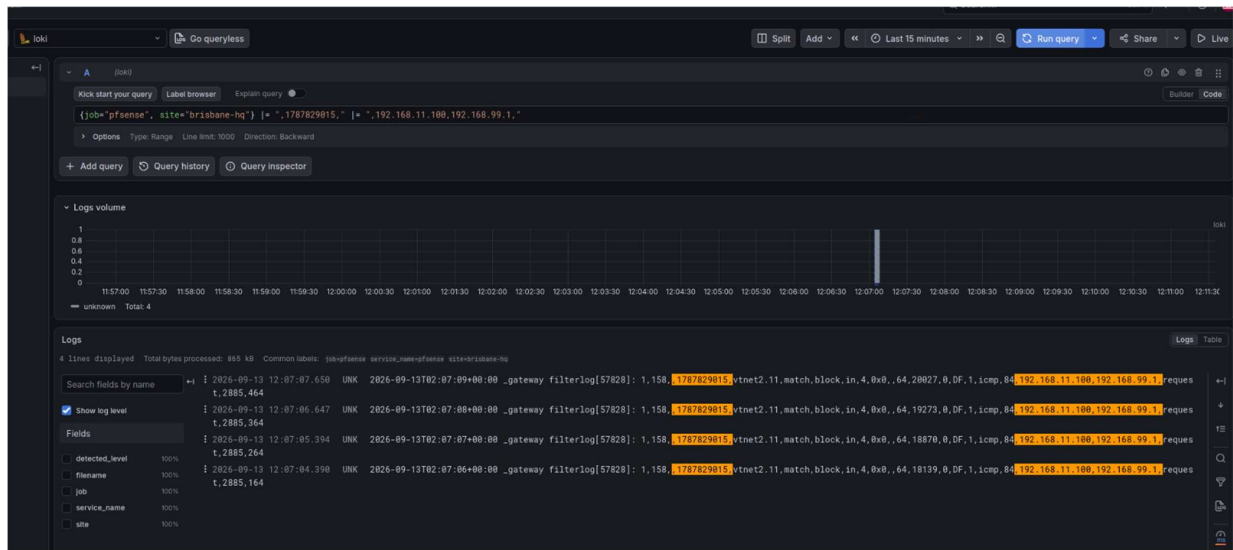
```
pc-admin@hq-emp-laptop: ~  
pc-admin@hq-emp-laptop:~$ date -u  
Sun Sep 13 02:06:42 AM UTC 2026  
pc-admin@hq-emp-laptop:~$ ping -c 4 -W 2 192.168.99.1  
PING 192.168.99.1 (192.168.99.1) 56(84) bytes of data.  
  
--- 192.168.99.1 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3110ms  
  
pc-admin@hq-emp-laptop:~$
```

Four ICMP requests to 192.168.99.1 received no replies. The corresponding firewall records provide the additional evidence linking this result to the tested rule.

Source evidence: evidence/T01\_Employee\_Ping\_Blocked.png

## Screenshot evidence continued

Figure E3 Correlated firewall records



Grafana retrieves four matching R06 records for source 192.168.11.100 and destination 192.168.99.1. This establishes the demonstrated collection and retrieval path.

Source evidence: evidence/T01\_Grafana\_R06\_Four\_Blocks.png